Response to P4043R0

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Abstract

P4043R0[1] ("Are C++ Contracts Ready to Ship in C++26?") asks EWG to reconsider the shipping vehicle for C++ Contracts. The paper cites 15 prior papers, raises no concern the committee has not already deliberated on, and proposes deferral as a remedy whose precondition - deployment experience - deferral itself prevents. Every argument in P4043R0[1] was before the committee when it adopted P2900R14[2] with strong consensus and when it subsequently rejected every proposal to reverse that decision.

Revision History

R0: March 2026 (pre-Croydon)

- Initial version.

1. Introduction

P4043R0[1] proposes three polls asking EWG to reconsider, defer, or change the shipping vehicle for C++ Contracts. The paper was posted on 2026-03-07, two weeks before Croydon, outside any official WG21 mailing.

The question this paper addresses is narrow: does P4043R0[1] present information or analysis that the committee has not already considered?

P4043R0[1] advances five claims:

1. The Contracts design continues to evolve ("design churn").
2. Papers raising concerns about Contracts have not reached consensus.
3. NB comments remain unresolved.
4. Deferral would allow implementation and deployment experience.
5. The committee should reconsider the shipping vehicle.

Claims 1 through 4 are offered as evidence for claim 5. Sections 2 through 6 examine each claim in order. Section 7 addresses procedural concerns.

The author has no involvement in the C++ Contracts facility and holds no position on its design.

2. Design Changes Since Adoption

Claim 1: The Contracts design continues to evolve.

P4043R0[1] characterizes post-adoption activity as evidence that "the design continues to evolve." The changes since adoption fall into three categories.

2.1 Routine NB Comment Resolutions

Two changes have been applied to the Working Draft since P2900R14[2] was adopted:

- P3819R0[4] removed `evaluation_exception()`. Three identical NB comments (NL 17.10, US 69-125, GB 04-124) identified the function as syntactic sugar that turned out to be unimplementable without significant complexity. EWG applied the fix at Kona 2025.
- P3886R0[5] added a feature-test macro that was missed. EWG applied the fix at Kona 2025.

Both are exactly the kind of fix the NB comment process exists to produce. Neither changes the design.

2.2 The Romanian NB Comment (RO 2-056)

NB comment RO 2-056 proposed new design for enforcement semantics. Seven of the 15 papers P4043R0[1] cites are entirely about this one comment:

Paper	Title
P3911R2[6]	Make Contracts Reliably Non-Ignorable
P3912R0[7]	Design considerations for always-enforced assertions
P3919R0[8]	Guaranteed-(quick-)enforced contracts
P3946R0[9]	Designing enforced assertions
P4005R0[10]	A proposal for guaranteed-(quick-)enforced contracts
P4009R0[11]	A proposal for solving all of the contracts concerns
P4015R0[12]	Enforcing Contract Conditions with Statements

The discussion was extensive. No design change resulted. The committee rejected each specific proposal; it did not poll against guaranteed enforcement as a goal. The outcome reinforced consensus for the adopted design.

2.3 Core Design

Unchanged. P2900R14[2] (published 2025-02-13) remains the latest revision. Zero design changes have been applied since adoption.

2.4 Assessment

P4043R0[1] conflates two routine NB fixes and one NB comment that generated extensive discussion but no design change with design instability. Two small fixes and a resolved NB comment are not churn. They are the NB comment process working as intended.

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3. Prior Deliberation on Each Concern

Claim 2: Papers raising concerns about Contracts have not reached consensus.

P4043R0[1] cites 15 papers. Every one was published before P4043R0[1]. The arguments they contain were before the committee in EWG telecons and at the Kona 2025 and Hagenberg 2025 meetings. The paper does not cite any new implementation result, deployment observation, or design analysis.

The following table maps each claim to the prior papers that raised it and the committee action that disposed of it:

Claim	Raised by	Disposition
1. Design continues to evolve	P3573R0[13], P3829R0[14], P3835R0[15]	See Section 2. Zero design changes since adoption.
2. Concern papers lack consensus	P3851R0[16], P3853R0[17], P3909R0[18]	Each proposal to modify or remove Contracts was rejected (P3846R0[3]).
3. NB comments remain unresolved	P3911R2[6], P3912R0[7], P3919R0[8], et al.	RO 2-056 is the one outstanding comment. P4043R0[1] proposes no solution.
4. Deferral enables deployment	P3909R0[18]	See Section 5. Deferral prevents the deployment it claims to enable.

Claim	Raised by	Disposition
5. Reconsider the shipping vehicle	EWG poll, 2025-02-11	Consensus against removal (P2899R1[19]).

P4020R0[20] ("Concerns about contract assertions") appeared in the 2026-02 pre-Croydon mailing and has not yet been discussed. It recommends no specific action beyond observing that "there are just 2 coherent actions" - keep Contracts or remove them. P4043R0[1] cites it but does not engage with its analysis.

A note on terminology: a committee poll produces one of three outcomes - consensus for, consensus against, or no consensus - and these are not interchangeable. Consensus against a proposal is an affirmative decision; no consensus means the committee did not reach a decision in either direction. P4043R0[1] states that proposals "have not yet achieved consensus" and that "none has yet reached consensus." The EWG poll on removing Contracts from C++26 was not "no consensus." It was consensus against removal. The proposals to modify the adopted design were not merely unsuccessful - they were rejected. Characterizing consensus against as the absence of consensus erases the committee's decision.

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4. Outstanding NB Comments

Claim 3: NB comments remain unresolved.

Of the NB comments related to Contracts, all have been resolved except RO 2-056. Section 2.2 documents the seven papers written in response to that comment and the outcome: extensive discussion, no design change, consensus for the current design reinforced.

P4043R0[1] cites the outstanding status of RO 2-056 as evidence of instability but proposes no solution to it. The paper recommends deferral or a White Paper - neither of which addresses the technical question RO 2-056 raises.

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5. Deferral and Deployment Experience

Claim 4: Deferral would allow implementation and deployment experience.

P4043R0[1] recommends deferral to allow "implementation and deployment experience." Language features do not receive widespread deployment outside standards. The Concepts TS (ISO/IEC TS 19217:2015[21]) demonstrated this: the TS was withdrawn, and Concepts were redesigned and shipped directly in C++20. Deferral does not produce deployment experience. Deferral prevents it.

The question of whether to ship Contracts as a TS rather than in the IS has been litigated. P3265R3[22] ("Ship Contracts in a TS") proposed exactly this. P3276R0[23] ("P2900 Is Superior to a Contracts TS") responded that no evidence supports the claim that a TS would resolve open questions, and that the established process for a TS requires listing specific goals, exit criteria, and a viable path through which answers will be acquired. The Direction Group addressed the same question in P4000R0[24] ("To TS or not to TS: that is the question"). SG21 polled on the question and reached consensus against shipping Contracts as a TS (P2899R1[19]).

P4043R0[1] suggests a White Paper as an alternative vehicle. A White Paper carries the same structural requirement as a TS: it must answer a clear set of questions through a viable process. P4043R0[1] identifies no such questions and proposes no such process.

`constexpr` shipped as a minimal language feature in C++11 (N2235[25]). C++14 relaxed its constraints dramatically (N3652[26]). The MVP model - ship a minimal feature, expand it in the next standard - is established WG21 practice.

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6. The Committee Record

Claim 5: The committee should reconsider the shipping vehicle.

On 2025-02-11, EWG polled on whether to remove P2900[2] ("Contracts for C++") from consideration for C++26 and find a different shipping vehicle. The result was consensus against removal (P2899R1[19]).

On 2025-02-13, LWG forwarded P2900R14[2] to plenary with consensus (P2899R1[19]).

On 2025-02-16, plenary adopted P2900R14[2] into the C++26 Working Paper with strong consensus (P2899R1[19]).

Since adoption, multiple proposals to modify the design have been presented; each was rejected (P3846R0[3]). The burden of proof lies with the paper proposing to reverse a decision taken with strong consensus. P4043R0[1] does not meet that burden.

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7. Procedural Concerns

P4043R0[1] (dated 2026-03-07) does not appear in any official WG21 mailing. The 2026-02 pre-Croydon mailing (released 2026-02-23) contains papers through P4032R0[27]. The paper was simultaneously posted to Reddit. The paper presents no proposed solution to the one outstanding NB comment (RO 2-056).

8. Conclusion

P4043R0[1] presents no new implementation result, no new deployment observation, and no new design analysis. The two post-adoption changes to the Working Draft are routine NB comment resolutions. The extensive discussion around RO 2-056 produced no design change and reinforced consensus for the adopted design. The alternative vehicles the paper suggests - C++29, a TS, a White Paper - have been considered and rejected, and P4043R0[1] provides no new basis for revisiting those decisions. The committee adopted P2900R14[2] with strong consensus. Nothing in P4043R0[1] warrants reversing that decision.

Acknowledgements

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References

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26. N3652 - "Relaxing constraints on constexpr functions" (Richard Smith, 2013). https://open-std.org/jtc1/sc22/wg21/docs/papers/2013/n3652.html
27. P4032R0 - last paper in the 2026-02 pre-Croydon mailing.